DEEP LEARNING MINI PROJECT • COMPREHENSIVE SUMMARY

Phishing URL Detection Using Deep Learning

Offline Classification System via Hybrid Subword N-Gram & Lexical Deep Neural Networks

DOMAIN:	ARCHITECTURE:	DATASET:	FEATURES / WEIGHTS:
Deep Learning & Cybersecurity	Hybrid N-Gram + Lexical DNN	246,755 URLs (Combined)	10,031 Features / 10,032 Weights

1. Executive Summary & Problem Statement

Phishing attacks remain one of the most prolific threat vectors in cybersecurity, deceiving users into surrendering sensitive credentials, payment details, and authentication tokens. Traditional countermeasures rely primarily on **static blacklists** (e.g. Google Safe Browsing) or **active web crawling**. However, blacklists suffer high vulnerability to zero-day ephemeral domains, live web crawling causes severe latency and connection timeouts, and transmitting every visited link to third-party verification servers leaks private browsing history.

Project Solution: An end-to-end **100% offline, privacy-preserving deep learning system**. The classifier extracts 10,031 hybrid features (10,000 subword character n-grams + 31 lexical, structural, and brand authority parameters) purely from the raw URL string in under 1 millisecond. The model operates without making any network requests, achieving a state-of-the-art **98.49% test accuracy** and **0.995 AUC-ROC**.

2. Technologies & Software Stack (What Is Used)

Technology	Role & Application in Project
TensorFlow & Keras	Builds the neural network computational graph, serializes model weights to my_model.keras, and runs accelerated tensor inference.
Scikit-Learn	Computes subword n-gram projections via FeatureHasher, scales feature distributions with MaxAbsScaler (scaler.pkl), and performs stratified evaluation (ROC-AUC, PR curves, Confusion Matrix).
Pandas & SciPy	Performs vectorized URL manipulation across 246,755 dataset rows and manages memory-efficient sparse CSR matrices for high-dimensional feature representations.
Flask & Web UI	Lightweight WSGI server providing interactive web interface (http://127.0.0.1:5000) and RESTful prediction endpoints (/api/predict).

3. Deep Learning Architecture (How It Is Used)

Layer	Type	Dimension	Activation	Function & Regularization
Input	InputLayer	10,031 Features	—	Sparse-aware MaxAbsScaler feature normalization
Subword Proj	FeatureHasher	10,000 Dimensions	Linear	Subword character n-grams (ranges 3 to 5)
Lexical Embed	Structural Parser	31 Features	Normalized	Domain, TLD, delimiter counts, brand authority flags
Output	Dense Layer	1 Neuron	Sigmoid	Calibrated Decision Threshold = 0.5000

- **Total Trainable Weights:** 10,032 parameters | **Loss:** Binary Cross-Entropy | **Regularization:** L2 Penalty (10^{-5})
- **Inference Latency:** < 1 ms per URL | **Model Size:** 55 KB serialized (my_model.keras).

4. Feature Engineering Pipeline (The 10,031 Hybrid Features)

Implemented in phishing_detector/features.py, raw URL strings are converted into 10,031 normalized features offline:

Feature Category	Features / Dims	Description & Security Significance
Subword N-Grams	`hash_0 ... hash_9999 (10,000 dims)`	Captures character n-grams (3-to-5 chars) spanning subdomain transitions, path tokens, keyword obfuscation, and subtle letter substitutions.
Structural & Host	`ip, punycode, port, ratio_digits_host, path_extension`	Detects raw IP hosts, homograph punycode spoofing (Unicode), abnormal ports, digit density, and dangerous script/executable extensions (.exe, .php).
Character Counts	`nb_hyphens, nb_at, nb_qm, nb_and, nb_eq, nb_underscore, nb_tilde, nb_percent, nb_colon, nb_comma, nb_dslash, ...`	Quantifies delimiter frequencies, credential '@' tokens, multiple query parameters, and double-slash obfuscations.
Subdomain & TLD	`tld_in_path, tld_in_subdomain, abnormal_subdomain, prefix_suffix, suspecious_tld`	Catches brand impersonation where genuine brand names reside inside subdomains (e.g. paypal.com.verify-user.tk) and suspicious TLDs (.xyz, .tk, .top).
Heuristics & Authority	`http_in_path, shortening_service, nb_redirection, phish_hints, is_trusted_brand_host`	Identifies link shorteners (bit.ly), protocol traps, deceptive keywords, and verifies legitimate brand domain registration boundaries.

5. Experimental Results & Evaluation Metrics

Trained on **80,000 balanced URLs** from combined_dataset.csv and evaluated on a held-out test split of **12,000 URLs** (15% partition):

98.49%	**0.995**	**0.996**	**0.5000**
TEST ACCURACY	AUC-ROC SCORE	PR-AUC (AVG PREC)	DECISION THRESHOLD

Confusion Matrix (Held-out Test Split = 12,000 URLs)

Ground Truth Class	Predicted Legitimate	Predicted Phishing	Precision	Recall	F1-Score
Legitimate (6,000)	**5,875** (True Neg)	125 (False Pos)	99.06%	97.92%	98.48%
Phishing (6,000)	56 (False Neg)	**5,944** (True Pos)	97.94%	99.07%	98.50%

6. End-to-End Inference Workflow & Interfaces

Workflow Pipeline: Raw URL → features.extract_features() [10,031 floats] → scaler.transform() → model.predict() → Sigmoid Probability p → Decision Rule ($p \ge 0.5000 \Rightarrow \text{Phishing}$) → Output Display.

- **Interactive Web Dashboard (Flask):** Run python -m web.app and access at http://127.0.0.1:5000. Provides dynamic probability gauges, visual confidence scores, and real-time verdicts.
- **CLI Utility:** Run single or batch URLs via python predict.py <url> with formatted text or --json output for automated SIEM pipelines.

7. Academic Syllabus & Course Outcomes Mapping

Syllabus Unit	Theoretical Concept	Implementation in Project
Module 1: Intro to DL	Motivation & Real-world Applications	Demonstrates deep learning's ability to learn non-linear patterns on zero-day cyber threats, overcoming static blacklist limitations.
Module 2: Neural Networks	High-Dimensional Mapping & Activations	10,031 input dimensions with non-linear Sigmoid activation mapping probability scores with binary cross-entropy loss.
Module 3: Optimization & Regularization	Gradient Optimization & Regularization	Stochastic Gradient Descent with log-loss; L2 regularization (10^{-5}); MaxAbsScaler feature standardization.